



Panchip Microelectronics Co., Ltd.

PAN3029/3060 Series Continuous Wave mode Application Reference

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PAN3029/3060 Series Single Carrier Application Reference

Documentation

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Revision History

Version	Revision Date	Description
V1.0	2023.06	Initial release
V1.1	2024.04	Modify chip description

Table of contents

1. Function Introduction	1
2 Software Design Reference	2
2.1 Software Design Process	2
2.2 Interface Description	2
2.3 Software Design Verification	2
2.3.1 SDK Examples	2
2.3.2 Verification Results	4



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1 Function Introduction

PAN3029/3060 series provides continuous wave (CW) test function. When configured in CW, the chip will transmit a CW for users. OLED routine mode D supports CW transmission test.

2 Software Design Reference

2.1 Software Design Process

1. Initialize the chip.
2. Configure CW related parameters and configure to enter CW mode.
3. Configure the frequency and set the chip to enter the CW transmission state.
4. If you want to modify the frequency, you need to wait for the TxDone IRQ signal, then exit the transmission state, reconfigure the frequency, and set it to enter the CW transmission state again.
5. If you want to exit the CW mode, you need to wait for the TxDone IRQ signal, then exit the transmission state and exit the CW mode.

2.2 Interface Description

```
RF_Err_t rf_set_carrier_wave_on(void);
```

Configure single carrier related parameters and configure to enter CW mode.

```
RF_Err_t rf_set_carrier_wave_freq(uint32_t freq);
```

Configure CW transmission frequency point and generate CW.

```
RF_Err_t rf_set_carrier_wave_off(void);
```

Configure to turn off CW mode and stop CW generation.

2.3 Software Design Verification

2.3.1 SDK Examples

Reference code:

```
ret = rf_init();//RF initialization
if(ret != OK)
{
    dis_err(" RF Init Fail");
    while(1);
}
```

```
}

uint32_t freq_now;

rf_set_carrier_wave_on();

#define REFQ_408    1//Band switch, open 408~565MHz traversal

#if REFQ_408

uint8_t i;

uint32_t freq_min = 408000000;  //starting frequency
for(i=0; i<158; i++)
{
    freq_now = freq_min + i*1000000; //Calculate the current frequency
    rf_set_carrier_wave_freq(freq_now); //Set frequency and start CW transmission
    while(rf_get_transmit_flag() != RADIO_FLAG_TXDONE)//Wait for TxDone IRQ
    {
        rf_irq_process();
    }
    rf_set_transmit_flag(RADIO_FLAG_IDLE); //Clear event flag
}

#else    //band 816~1080MHz

uint16_t j;

uint32_t freq_min = 816000000;
for(j=0; j<265; j++)
{
    freq_now = freq_min + j*1000000;
    rf_set_carrier_wave_freq(freq_now);
    while(rf_get_transmit_flag() != RADIO_FLAG_TXDONE)//Wait for TxDone IRQ
    {
        rf_irq_process();
    }
}
```

```

}
rf_set_transmit_flag(RADIO_FLAG_IDLE);
}
#endif

rf_set_carrier_wave_off();//Stop CW transmission
while(1){}

```

The sample code configures the output of 408~565MHz and 816~1080MHz single carrier output, and switches the program through the macro REFQ_408.

The user can select the frequency range of CW as needed. If rf_set_carrier_wave_off() is not executed, the chip will continue to output CW at the current frequency.

2.3.2 Verification Results

The spectrum analyzer display results are shown in Figure 1-1:



Figure 1-1 Spectrum analyzer display results